

Insights & Recommendations

Findings from the Power BI Dashboard — Airline Flight Delay Analysis

Data Source	Maven Analytics — Airline Flight Delays (U.S. DOT, 2015)
Basis	Power BI dashboard analysis, 5,819,079 flights
Version	1.0
Date	August 2026

1. Executive Summary

Our analysis covers **5.82 million flights** from 2015. Under our project definition, **2.09 million flights were delayed**, giving a **35.9% delay rate**, while **89,884 were cancelled**. Flight volume was relatively stable across the year, but delay rates varied significantly by month and airline. The largest contributors to delay minutes were **late aircraft, airline-related factors, and air-system delays**. In contrast, **weather was the dominant cancellation driver**, accounting for 54.35% of cancellations. This suggests that reducing delays requires a focus on operational efficiency and delay propagation, while cancellation management should place greater emphasis on weather preparedness and contingency planning. High-volume airports and high-delay-rate airlines should be investigated further using normalized rates rather than raw counts alone.

2. Methodology Note

All figures below are read directly from the Power BI dashboard's validated KPI cards and visuals, not from Power BI's abbreviated/rounded card display values. Findings are labeled as **analytical observations**, not causal claims — the dataset shows correlation and concentration patterns, not proven cause-and-effect relationships. Wherever a finding could be confounded by flight volume (e.g., an airport having more delayed flights simply because it operates more flights overall), that limitation is called out explicitly and a normalized follow-up metric is recommended.

Part 1 — Business Insights

1.1 Overall Operational Performance

Metric	Value
Total Flights	5,819,079
On-Time Flights	3,627,112
Delayed Flights	2,086,896
Cancelled Flights	89,884
Diverted Flights	15,187
Overall Delay Rate	35.9%

Finding: The operation handled an extremely large volume of flights, but delays affected more than one-third of the flights in scope. **Business interpretation:** This makes delay management the primary operational issue identified by the dashboard.

1.2 Monthly Flight Performance

Monthly flight volume remained relatively stable throughout 2015, generally in the range of 0.43M-0.52M flights per month. Delay rate, however, fluctuated considerably:

- **February and June** were among the highest-delay-rate months, at roughly 41%.
- **September and October** were among the lowest, at roughly 29%.
- **December** showed a noticeable recovery in delay rate.

Finding: Delay performance varied more substantially than overall flight volume, meaning monthly delay-rate swings cannot be explained by changes in flight volume alone. **Business interpretation:** Operational performance appears to be influenced by factors beyond flight count — seasonality, weather, aircraft rotations, airline operations, and air-system constraints should be investigated when planning operational capacity. *This is an analytical observation, not a causal claim — the dashboard does not establish a causal relationship between month and delay.*

1.3 Airline Performance

The Top 5 Airlines by Delay Rate are:

Rank	Airline (IATA)
1	NK
2	F9
3	HA
4	VX
5	US

The leading carriers in this group reach delay rates of roughly 40-48%.

Finding: Delay performance varies substantially between airlines, with the highest-delay-rate carriers operating materially above the lower end of the Top 5 group. **Business interpretation:** Airline-level operational practices may warrant further investigation, including aircraft turnaround processes, scheduling buffers, aircraft rotation, crew availability, maintenance-related constraints, and route/network structure. *These factors are potential areas for investigation, not confirmed causes — the dashboard does not directly prove causation.*

1.4 Airport Concentration

The Top 5 Origin Airports by Delayed Flights are:

Rank	Airport (IATA)
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1	ATL
2	ORD
3	DFW
4	DEN
5	LAX

ATL and ORD each exceed roughly 100,000 delayed flights.

Finding: A relatively small group of high-volume origin airports accounts for a substantial concentration of delayed flights. **Business interpretation:** These airports should be treated as priority locations for operational investigation.

Analytical caveat: High delayed-flight volume does not necessarily mean poor airport performance — a high-volume airport can naturally show more delayed flights simply because it handles more flights overall. A follow-up analysis should compute **Delayed Flights ÷ Total Flights at Airport** to determine each airport's true delay rate, rather than relying on raw counts alone.

1.5 Delay Severity

Delay Category	Share of Delayed Flights
1-15 minutes	50.96%
16-30 minutes	18.73%
31-60 minutes	15.03%
61-120 minutes	9.81%
120+ minutes	~5.47%

Finding: More than half of delayed flights (50.96%) fall into the shortest delay category, but a meaningful portion — roughly 49.04% — experience disruptions longer than 15 minutes. **Business interpretation:** Not every delay carries equal business impact; a 5-minute delay and a 2-hour delay have very different consequences. The longer-duration categories deserve particular attention, as they can create passenger disruption, missed connections, aircraft-rotation problems, crew-scheduling problems, and downstream delays.

1.6 Root Cause of Delay

This is one of the strongest findings in the project. The dashboard shows the largest contributors to total delay minutes, ranked highest to lowest:

1. Late Aircraft
2. Airline (Air Carrier)
3. Air System (NAS)
4. Weather
5. Security

This pattern is consistent with the broader U.S. Bureau of Transportation Statistics (BTS) figures for 2015, which show approximately 39.8% of delay minutes attributable to late-arriving aircraft, 32.2% to air carrier, 22.9% to national aviation system, 4.9% to extreme weather, and 0.1% to security.

Finding: Late aircraft, airline-related factors, and air-system delays account for the overwhelming majority of delay minutes, while weather and security represent much smaller shares. **Business interpretation:** The biggest opportunity is not simply weather mitigation — the data indicates that most delay minutes are associated with operational/system factors that may be more actionable than weather or security disruptions.

1.7 Cancellation Analysis

Total cancellations: **89,884**. The cancellation-reason breakdown is:

Cancellation Reason	Share of Cancellations
Weather	54.35%

Airline / Carrier	28.11%
National Air System	17.52%
Security	Smallest share

Finding: Weather is the dominant cancellation driver, accounting for more than half of all cancellations — a very different pattern from the delay-cause analysis. **Key insight:** weather is relatively minor in total delay minutes but dominant in cancellations. This means weather may not be the largest source of accumulated delay time, but when weather disruption does occur, it has a much stronger relationship with a flight being cancelled outright rather than merely delayed.

1.8 Delay vs. Cancellation — Key Strategic Finding

Main Driver(s)	
Delays	Late Aircraft → Airline → Air System
Cancellations	Weather

Business conclusion: The operational response should differ by disruption type. For **delays**, the focus should be on operational efficiency and network/system performance. For **cancellations**, the focus should be heavily on weather preparedness, contingency planning, and disruption management. This distinction — that the drivers of delay and the drivers of cancellation are fundamentally different — is one of the strongest overall insights from the dashboard.

Part 2 — Business Recommendations

The following recommendations translate the findings in Part 1 into actionable next steps.

Recommendation 1 — Prioritize Late-Aircraft Operations

Evidence: Late aircraft is the largest contributor to total delay minutes.

Recommendation: Investigate aircraft turnaround time, aircraft rotation schedules, buffer times between flights, recurring aircraft-level delay patterns, and delays propagating from earlier flights.

Expected objective: Reduce delay propagation across subsequent flights.

Recommendation 2 — Target High-Delay Airlines

Evidence: The Top 5 airline visual shows substantial differences in delay rates.

Recommendation: Conduct a deeper airline-level performance review covering delay rate, average delay duration, cancellation rate, delay causes, and airport/route mix.

Important: Do not penalize an airline simply because it appears in the Top 5 — compare performance after accounting for flight volume and operating mix.

Recommendation 3 — Prioritize High-Volume Airports

Evidence: ATL, ORD, DFW, DEN, and LAX have the highest delayed-flight volumes.

Recommendation: Prioritize these airports for deeper operational analysis, but base the next round of analysis on **Delayed Flights ÷ Total Flights** rather than raw delayed-flight counts.

Why it matters: This separates "this airport has many delays because it is huge" from "this airport has an unusually high rate of delays."

Recommendation 4 — Separate Minor and Major Delays

Evidence: 50.96% of delayed flights fall into the 1-15 minute category, while approximately 49% exceed 15 minutes.

Recommendation: Create separate operational targets for minor, moderate, and severe delays. Management should especially monitor the 61-120 minute and 120+ minute categories.

Recommendation 5 — Build Weather Contingency Planning

Evidence: Weather accounts for 54.35% of cancellations in the dashboard.

Recommendation: Strengthen weather disruption forecasting, proactive passenger communication, rebooking procedures, standby capacity, and aircraft/crew contingency planning.

Goal: The aim is not to eliminate weather disruptions — that isn't possible — but to reduce their operational and passenger impact when they occur.

Recommendation 6 — Monitor Air-System Constraints

Evidence: Air-system delays are the third-largest contributor to delay minutes.

Recommendation: Investigate the airports, routes, and time periods where air-system delays concentrate, and coordinate capacity planning around known congestion periods.

Recommendation 7 — Establish a Recurring Operational Monitoring Dashboard

This dashboard should not remain a one-time report. A production version should track Delay Rate, Cancellation Rate, Average Delay, Severe Delay %, airline performance, airport performance, delay cause, month/season, and route, with monthly or weekly refreshes.

3. The Final Executive Story

Our analysis covers **5.82 million flights** from 2015. Under our project definition, **2.09 million flights were delayed**, giving a **35.9% delay rate**, while **89,884 were cancelled**. Flight volume was relatively stable across the year, but delay rates varied significantly by month and airline. The largest contributors to delay minutes were late aircraft, airline-related factors, and air-system delays. In contrast, weather was the dominant cancellation driver, accounting for 54.35% of cancellations. This suggests that reducing delays requires a focus on operational efficiency and delay propagation, while cancellation management should place greater emphasis on weather preparedness and contingency planning. High-volume airports and high-delay-rate airlines should then be investigated further using normalized rates rather than raw counts alone.